EDITS: Enhancing DIgital TwinS with scalable and interpretable AI

2nd meeting, 16/03/26

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ULB Machine Learning Group
WEL-RI investigator

Digital twin (DT)

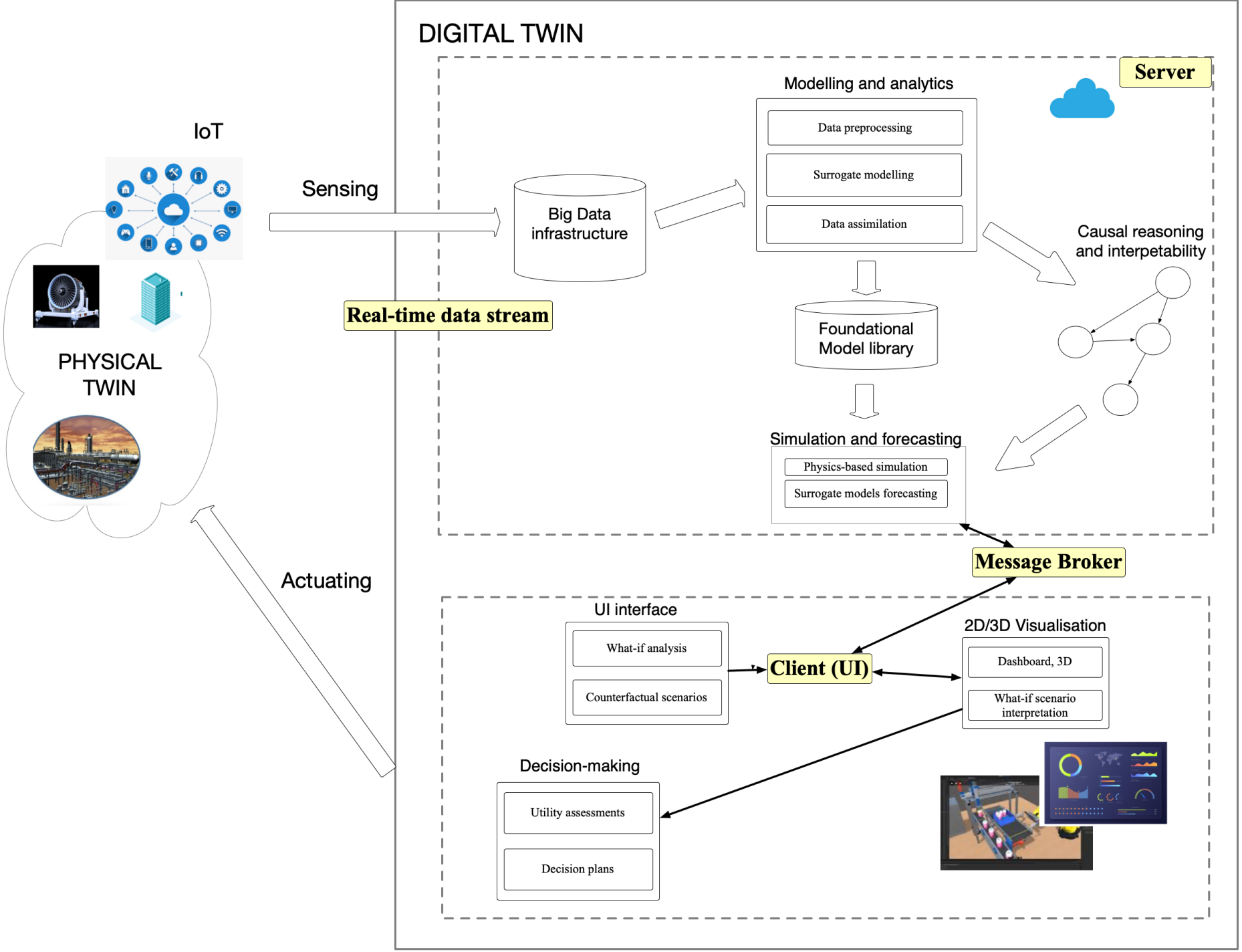
- Digital avatar of an object/system/process called **physical twin** (PT),
- A DT aims to
  - emulate,
  - explain,
  - anticipate,
  - control

the behaviour of the PT by relying on

  - multiscale physics-based models
  - data-driven ML models

whose accuracy, up-to-dateness and validity are guaranteed by a **continuous two-way flow of data** (measures and actions) between the DT and the PT.

The vision


DIGITAL TWIN
Server
IoT
Sensing
Real-time data stream
PHYSICAL TWIN
Actuating
Big Data infrastructure
Modelling and analytics
Data preprocessing
Surrogate modelling
Data assimilation
Foundational Model library
Simulation and forecasting
Physics-based simulation
Surrogate models forecasting
Causal reasoning and interpetability
Message Broker
UI interface
What-if analysis
Counterfactual scenarios
Client (UI)
2D/3D Visualisation
Dashboard, 3D
What-if scenario interpretation
Decision-making
Utility assessments
Decision plans

Application domains

- Energy
  - Grid
  - Transformers
  - Wind turbines
- Architecture and building
- Climate and weather
- Industrial plants: Industry 4.0
- Applied sciences: Fluidodynamics, combustion plants
- Smart cities
  - Mobility
  - Roofs
  - Trees

DT and the convergence of disciplines

- AI
- Modelling
- Simulation (what-if scenarios)
- Data analysis and machine learning
- Causal inference (interpretability, counterfactuals)
- Big data infrastructures (streaming data)
- Control
- Decision-making

Beyond the buzzword… hot scientific questions

- Integration of physical and data-driven models
  - Data assimilation
  - Physically based machine learning
  - Surrogate modelling
  - Causal inference and discovery
- Optimisation
  - Multi-criteria
- ML
  - Transfer learning, active learning
  - Continual learning
    - Non-stationarity, Concept drift
- Data infrastructure
  - big data
  - streaming infrastructure
  - Sequential model updating
  - simulation tools and languages
- User interaction
  - What-if scenario
  - Dashboarding/visualisation

EDITS workplan

ManMonths		10/10/25	10/04/26	10/10/26	10/04/27
	Man-Months	M1-M6	M7-M12	M13-M18	M19-M24
WP1- BD infrastructure (M1-M48)	**24**	**PD1**	**PD1, PD2**	**PD2**	**PD2**
WP2 -Calibration of the physics-based model with large variate data streams and integration with surrogate ML models (M4-M33)	**27**	**PD1,PHD**	**PD1,PHD**	**PD1,PD2, PHD**	**PD1, PD2,PHD**
WP3- DT Foundational models and transfer learning (M19-M30)	**12**				**PHD**
WP4- Causal discovery and inference (M13-M48)	**24**			**PHD**	**PHD**
WP5- Forecasting and simulation (M13-M42)	**24**			**PD1,PD3**	**PD3**
WP6- Case studies (M1-M48)	**36**	**PD1**	**PD1, PD2**	**PD2**	**PD2, PD3**

- Starting date: 1/1/26
- We are in M3
- 1st report on 31/1/26

Personnel

- Eladio MONTERO PORRAS
  - postdoc,
  - Digital twin for mobility, SUMO mobility simulator
  - **1/1/26-> 31/12/26**, 12 months
- Carlo FABRIZIO
  - PhD student
  - **1/2/26->**, 48 months
- Gianmarco PALDINO
  - Got a PhD on 4/3/26 in TRAIL/ARIAC project
  - Forecasting, causal inference, digital twins in energy domain
  - **1/5/26->**…
- Cédric SIMAR
  - Currently PhD student and teaching assistant
  - Signal processing, deep learning, robotic hand
  - **1/10/26->**…
- MLG but non WEL-T funded:
  - Natalia GARCIA COLIN, Business developer
  - Martin COLOT, Teaching assistant
  - Pascal TRIBEL, Teaching assistant
  - Alejandro MORALES, postdoc on Interreg project
  - Alexandra SEBASTIEN, not funded
  - Several MA students

WPs

1. Data Infrastructure (Paldino)
   - Sensing
     - Real data
     - Synthetic data
   - Big data streaming: (Fabrizio)
   - Data collection and storage
     - Database (Fabrizio)
     - Big data architecture
     - Cloud server
   - Dashboarding
   - Orchestration (Paldino)
2. Calibration and Learning (Fabrizio)
   - Continual/online
   - Batch/offline
   - Data assimilation
   - Integration with physical models (Tribel)
3. Foundational models and transfer learning (Simar)
4. Causality (Paldino)
   - Counterfactuals
   - interpretability
5. Forecasting and Simulation (Paldino, Montero)
   - What-if scenarios
   - User interaction
   - Visualisation
6. Case studies definition
   - Mobility: Traffic Twin (Montero)
   - Energy: VPP (Paldino)
   - EMG robotic hand (Simar)
   - Toy case: DT of a plant (MA thesis)

1st report (31/1/26)

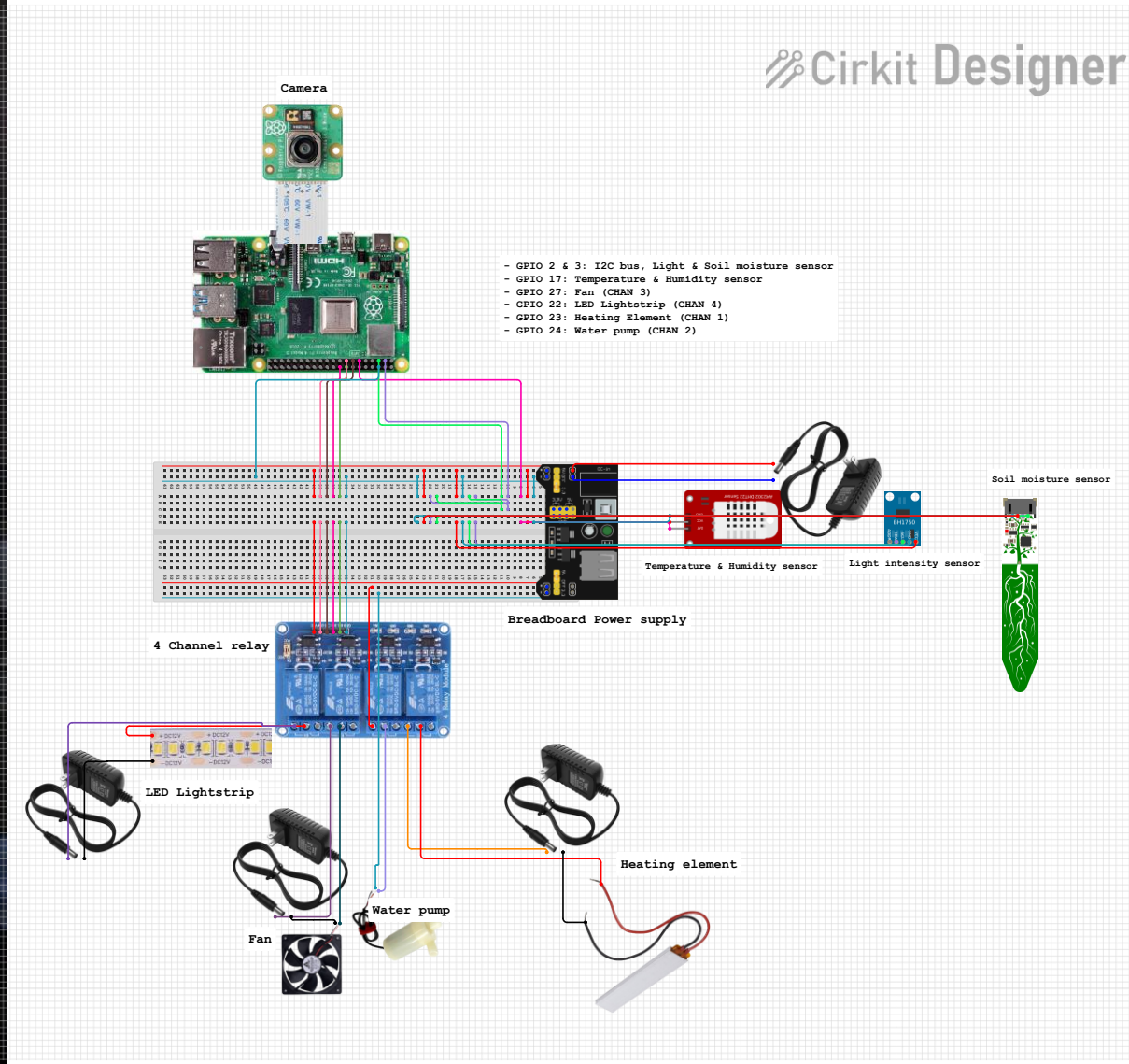
- 2 journal publications
- 2 scientific meetings
- 1 POC funding

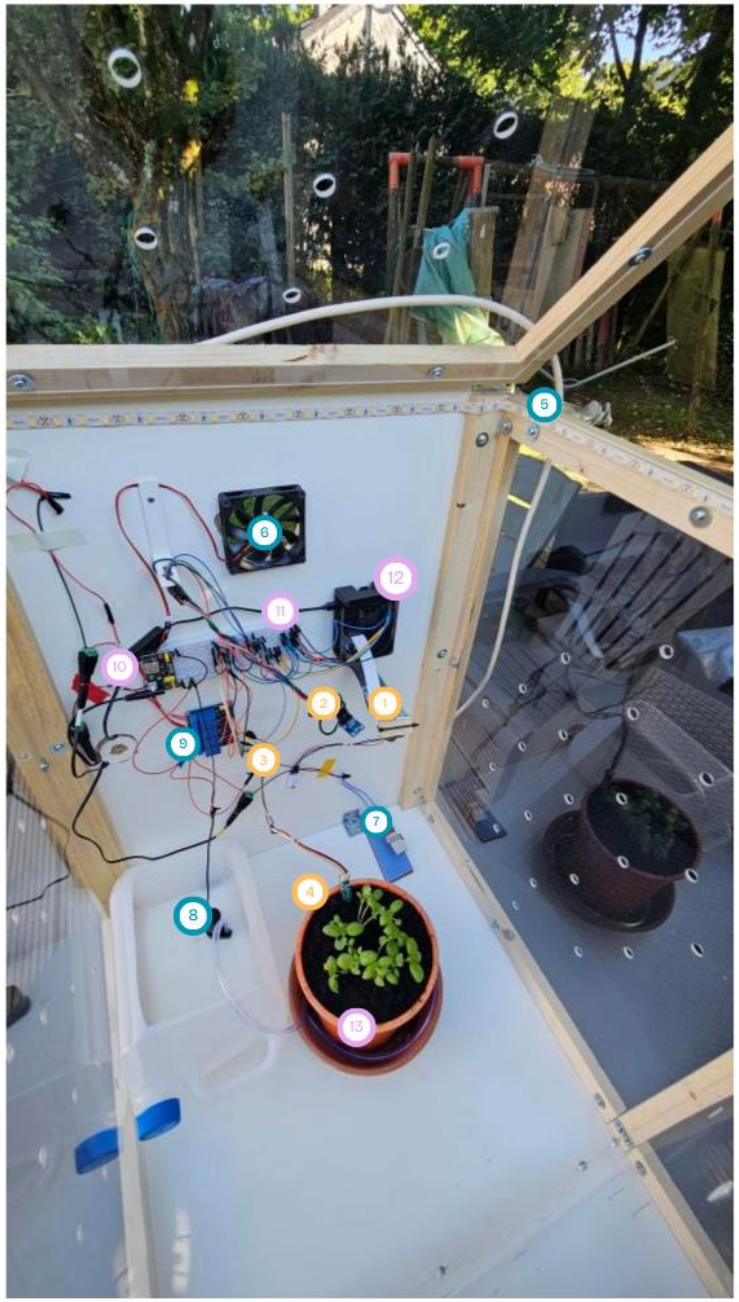
For the next report

- Journal publications
  - Paldino, Gian Marco; Bontempi, Gianluca "Causal Discovery in Multivariate Time Series through Mutual Information Featurization", International Journal of Forecasting, 2026, to appear, IF: 7.1
  - Paldino, Gian Marco; Caelen, Olivier; Oueslati, Marouene; Ansay, Marc; Johanesa, Tojo Valisoa A; Bontempi, Gianluca "On the integration of Domain Adaptation and Causal Discovery in Digital Twins: A Case Study for Plastic Injection Molding", Future Generation Computer Systems,108427,2026,North-Holland, IF: 6.1
  - Morales-Hernández A. , De Caro Fabrizio, Paldino Gian Marco, Tribel Pascal, Vaccaro Alfredo, Bontempi, Gianluca "A Direct Classification Approach for Reliable Wind Ramp Event Forecasting under Severe Class Imbalance", Electric Power Systems Research, to appear, IF: 4.2
- Conference publications:
  - Verhelst, Théo; Bontempi, Gianluca "Identifying counterfactual probabilities using bivariate distributions and uplift modelling", ESANN conference, 2026
- Events
  - Seminar in LIDAM, UCL, LLN, February, 2026
  - Keynote at Belgian AI Week, "Digital Twins and AI: the science beyond the hype", March 2026
  - Invited teacher at "Summer school on Machine Learning and Econometrics", SoFIE, Torino (I), June 2026
  - Invited teacher at EABCN training class, Nice, October, 2026
- Others
  - Interview Le Vif, KompasMedia BE about predictive maintenance, March 2026
  - Responsible for FARI Cities Flagship initiative
- Invited talks about DTs in MLG
  - Miguel Mendez, VKI, B, "Reinforcement Twinning: From Digital Twins to Acting Models in Thermo-Fluid Engineering"
  - Alberto Procacci: "Reduced-order surrogate models of physical systems: physics-constraints and temporal forecasting", ULB, Polytech

Outline of presentations

1. Gianmarco PALDINO: the EDITS infrastructure
2. Carlo FABRIZIO: intelligent databases for streaming data
3. Eladio MONTERO: the Ttwin case study
4. Gianmarco PALDINO: the Energy case study
5. Cedric SIMAR : the robotic hand case study
6. Natalia GARCIA: perspectives of dissemination and valorization


Cirkit Designer
Camera
- GPIO 2 & 3: I2C bus, Light & Soil moisture sensor
- GPIO 17: Temperature & Humidity sensor
- GPIO 27: Fan (CHAN 3)
- GPIO 22: LED Lightstrip (CHAN 4)
- GPIO 23: Heating Element (CHAN 1)
- GPIO 24: Water pump (CHAN 2)
Soil moisture sensor
Temperature & Humidity sensor
Light intensity sensor
Breadboard Power supply
4 Channel relay
LED Lightstrip
Heating element
Water pump
Fan



1. Camera
2. Light intensity sensor
3. Temperature & Humidity sensor
4. Soil moisture sensor
5. LED Lightstrip
6. Fan
7. Heating element
8. Water pump
9. 4 Channel relay
10. Breadboard power supply
11. Breadboard
12. Raspberry PI
13. Basil (Physical Twin)

Sensors

Actuators

Structure